Aadesh Mahesh Khande

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SUMMARY

Third-year B.Tech Computer Engineering (Software Engineering) undergraduate who shipped production features across four multi-tenant SaaS platforms during a three-month SDE internship at CustomSoft.

EXPERIENCE

CustomSoft, Pune *Jun 2026 – Sep 2026*

Software Development Engineer (SDE) Intern

- Delivered production features across four multi-tenant SaaS platforms — healthcare, pet grooming, vehicle logistics, and field service — owning work end to end from schema design to cloud deployment.
- Eliminated double-booking races with atomic MongoDB findOneAndUpdate writes and distributed Redis mutex locks (SET NX EX, Lua release), holding checkout under 150 ms at 50 concurrent users.
- Built a gateway-agnostic payment layer with Razorpay HMAC-SHA256 webhook verification and replay protection, splitting every payment across platform, staff, and salon ledgers in minor units.
- Built a context-injected AI layer (Groq Llama 3.1-8B, Anthropic Claude) that auto-triages complaints, prescribes from a 50+ item exercise library, and drafts support replies.
- Engineered a recurring-job engine that creates its next instance inside a MongoDB transaction, hardened six-role RBAC with TOTP MFA and WebAuthn passkeys, and queued receipts on BullMQ workers.

PROJECTS

ThermalWatch AI — Satellite Thermal Intelligence (SIH 2026, Team Project) *Aug 2026 – Sep 2026*

Python, PyTorch, XGBoost, scikit-learn, Geospatial, Firebase, React **Live Demo**

- Fused nine satellite and ground telemetry streams (NASA FIRMS, ISRO INSAT, JAXA Himawari-9, ESA Sentinel, ECMWF ERA5, CPCB/OpenAQ) into a 2.2 million-observation training corpus.
- Eliminated geospatial train-test leakage with spatial block hashing over 0.25-degree tiles, and handled a 71.4% class skew with cost-sensitive focal loss.
- Fused XGBoost, a 1D-CNN over 144-step diurnal radiance curves, and a ResNet-18 on 10 m imagery into a stacking meta-learner reaching 88.17% zero-leakage macro F1.
- Ran a forensic audit that rejected a 655 MB synthetic imagery set and an impossible 303,318 K sensor reading, keeping the corpus to verified physical telemetry.

Rewind — Cross-Browser Playback Sync Extension *Apr 2026 – Sep 2026*

JavaScript, Chrome MV3, Firefox MV2, MutationObserver, React, Vite, Firestore **Live Demo**

- Published a cross-browser extension for Chrome (Manifest V3 service worker) and Firefox (Manifest V2 with explicit CSP whitelists) that syncs playback timestamps across Netflix, YouTube, and Crunchyroll.
- Tracked SPA video mounts with MutationObserver and WeakSet so references self-release during garbage collection, with a 30-second checkpoint and beforeunload hook to survive tab closes.
- Designed a dual-bucket Firestore pipeline in which the authenticated dashboard validates buffered payloads and stamps server-atomic timestamps before they reach permanent history.

EDUCATION

MIT Academy of Engineering, Pune *Sep 2024 – Present*

Bachelor of Technology in Computer Engineering (Software Engineering) **CGPA: 8.0**

Modern College of Arts, Science and Commerce *Sep 2022 – Feb 2024*

Class XII — Higher Secondary Certificate (HSC) **81.67%**

Pune Police Public School *2022*

Class X — Secondary School Certificate (SSC) **93.40%**

SKILLS

- **Languages:** Python, Java, C++, C, JavaScript, TypeScript, SQL
- **Frontend:** React, Next.js, Vite, Tailwind CSS, Framer Motion, Three.js, Recharts
- **Backend, Data & ML:** PyTorch, XGBoost, scikit-learn, Node.js, Express, REST APIs, MongoDB, PostgreSQL, Firebase, Redis, BullMQ
- **Tools:** Git, GitHub, Vercel, Render, Razorpay, Cloudinary, Zod, Jest, Vitest

INTERESTS

- Indie and AAA Games, Open-Source Software, Music, Chess, Touch Typing.